WEEK 4 – POWER BI

Time-Series Visualization & Financial Dashboard

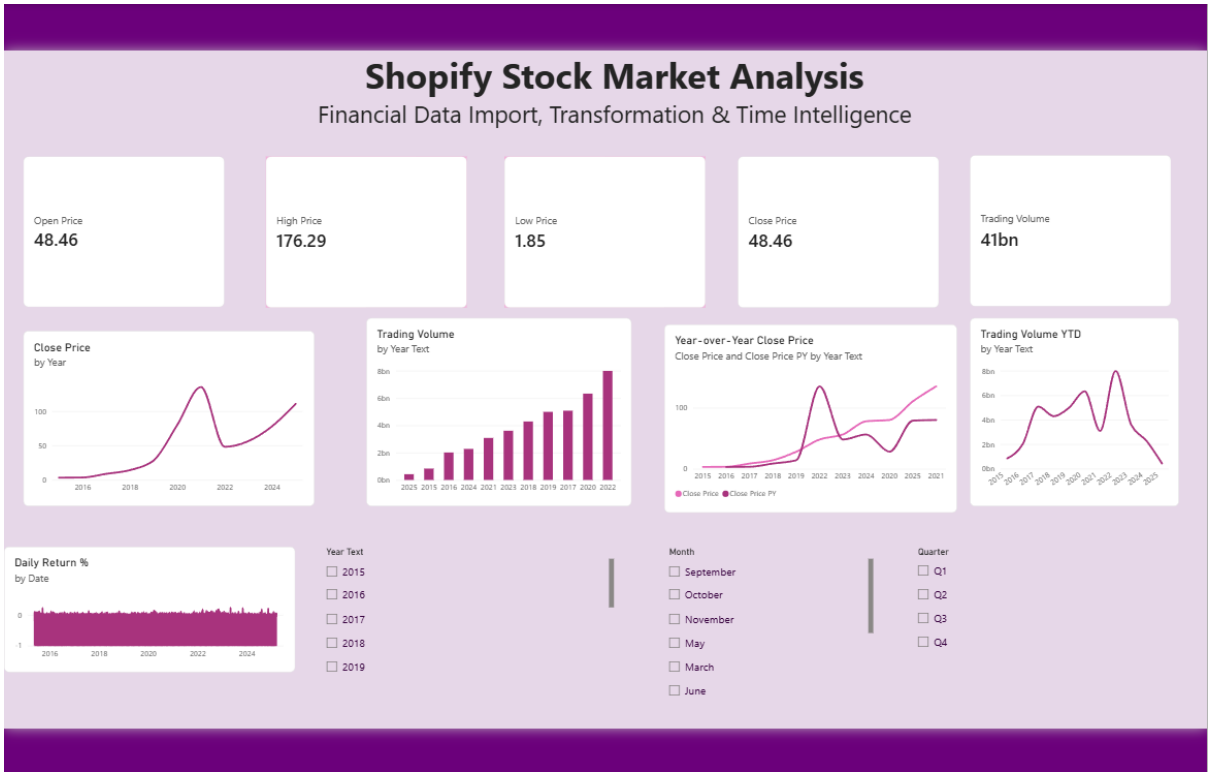
-done by **Sai Meera R** from **B.sc Data Science & AI**

Overview of Week 3:

In Week 3, we worked with the financial stock dataset in Power BI and prepared it for analysis. The data was imported and transformed, and a Date Table was created for time-based analysis. Measures such as Open Price, Close Price, High Price, Low Price, Trading Volume, Daily Return %**,** Close Price PY, Close Price YoY %, and Trading Volume YTD were created using DAX.

A financial dashboard was then created using cards and charts to understand stock price and trading-volume trends over time. This Week 4 task continues from the Week 3 dashboard by adding Exploratory Data Analysis (EDA), moving averages, interactive filters, field parameters, and custom tooltips.

Week 3 Financial Data Analysis Dashboard:



Introduction:

The Week 4 task focuses on Exploratory Data Analysis (EDA) using Power BI. The main objective is to analyze stock price and trading volume data using moving averages, crossover signals, interactive visualizations, slicers, and custom tooltips.

The analysis helps in understanding price trends, trading volume patterns, and changes in stock performance over time.

Objectives:

The main objectives of this task are:

- To calculate 20-Day Simple Moving Average (SMA).
- To calculate 50-Day Simple Moving Average (SMA).
- To create a Moving Average Crossover Signal.
- To visualize stock price trends using interactive charts.
- To analyze Trading Volume and Volume 20-Day SMA.
- To create interactive date slicers.
- To use a Field Parameter for selecting different price measures.
- To create a custom tooltip page for detailed stock information.

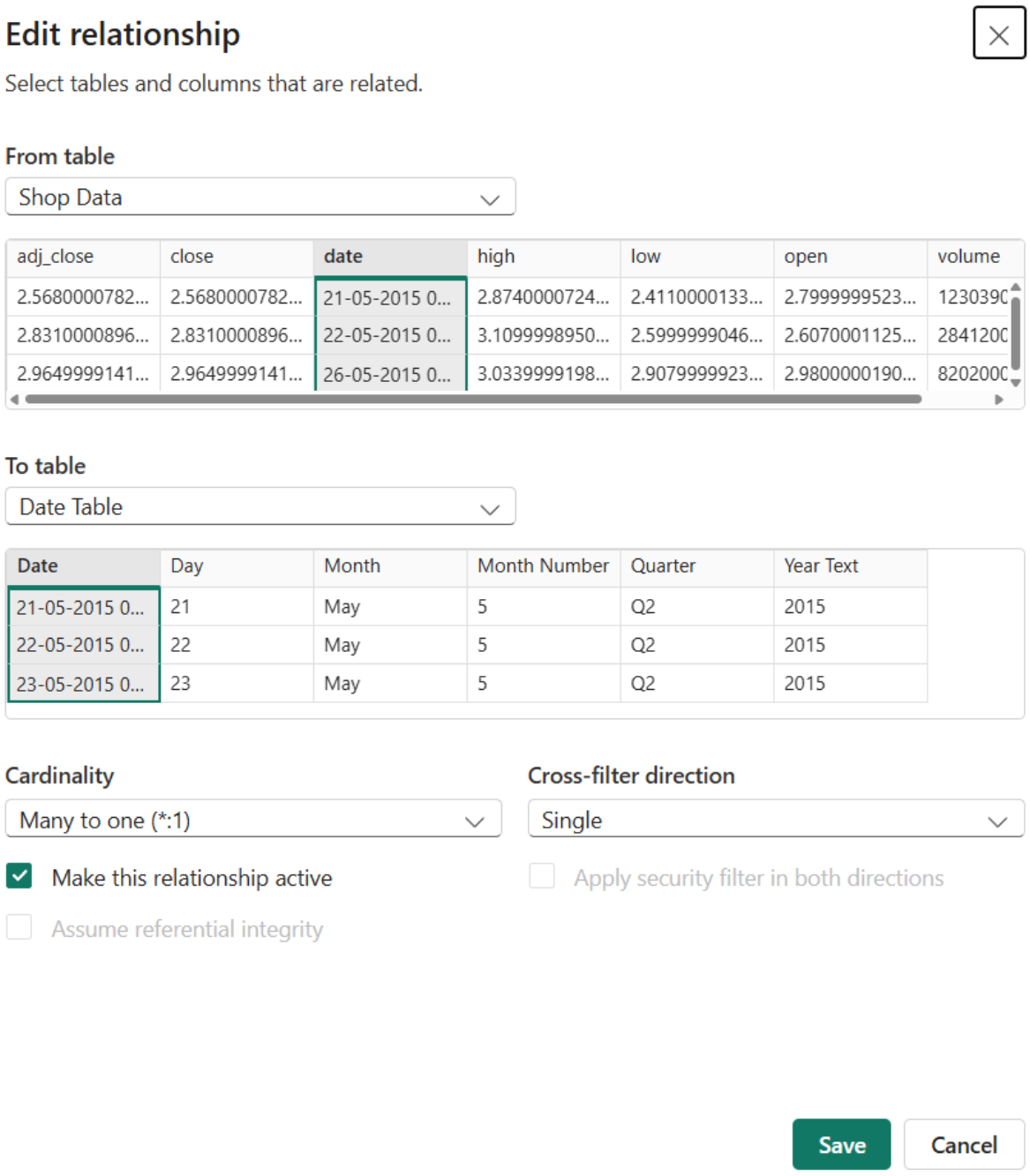
Dataset Used

The dataset used for this analysis is Shop Data, which contains stock-related information such as:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

A separate Date Table was used for date-based analysis and time intelligence.

Data Model with Shop Data and Date Table:



Data Analysis Using Moving Averages

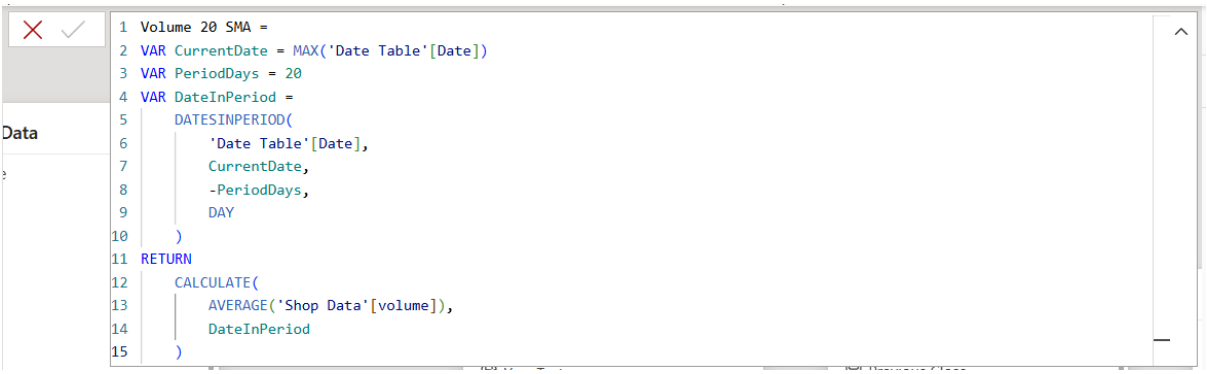
20-Day Simple Moving Average

A 20-Day SMA was created using DAX to calculate the average value over the previous 20 days.

The measure uses DATESINPERIOD() to identify the required date period and AVERAGE() to calculate the moving average.

Purpose:
The 20-Day SMA helps to observe short-term trends in the stock data.

DAX Measure for Volume 20-Day SMA



```
Volume 20 SMA =
VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 20
VAR DateInPeriod =
    DATESINPERIOD(
        'Date Table'[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
    CALCULATE(
        AVERAGE('Shop Data'[volume]),
        DateInPeriod
    )
```

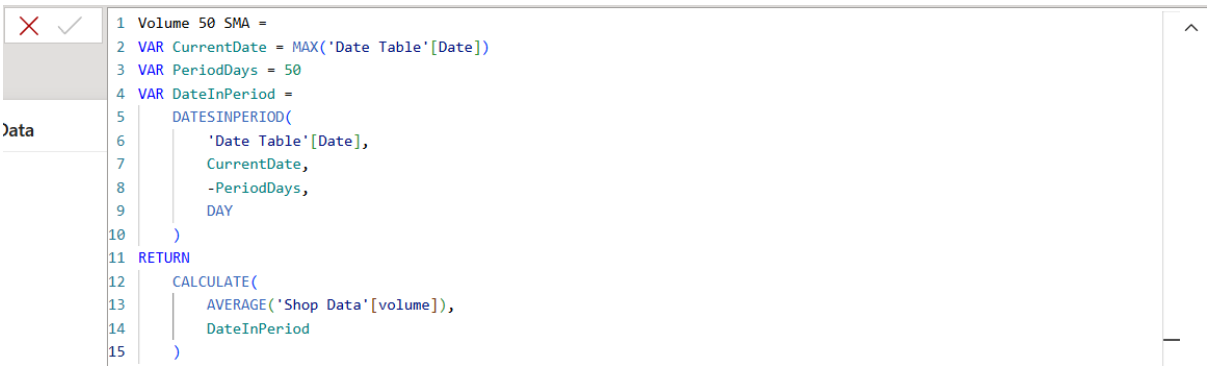
50-Day SMA

A 50-Day SMA was created to observe the longer-term trend.

The calculation uses a 50-day period and calculates the average value over that period.

Purpose:
The 50-Day SMA provides a broader view of the stock trend compared with the shorter 20-day period.

DAX Measure for 50-Day SMA



```
Volume 50 SMA =
VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 50
VAR DateInPeriod =
    DATESINPERIOD(
        'Date Table'[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
    CALCULATE(
        AVERAGE('Shop Data'[volume]),
        DateInPeriod
    )
```

Moving Average Crossover Signal

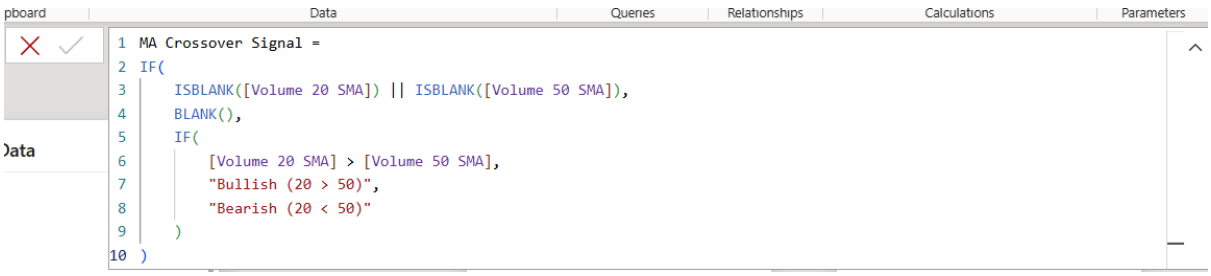
A Moving Average Crossover Signal was created using the 20-Day SMA and 50-Day SMA.

The measure compares the two moving averages:

- When the 20-Day SMA is greater than the 50-Day SMA, the signal is displayed as Bullish (20 > 50).
- When the 20-Day SMA is lower than the 50-Day SMA, the signal is displayed as Bearish (20 < 50).

This provides an easy way to observe the relationship between the short-term and longer-term moving averages.

Moving Average Crossover Signal



Interactive Price Trend Visualization

A Line and Stacked Column Chart was created to visualize the stock price trend.

The chart uses:

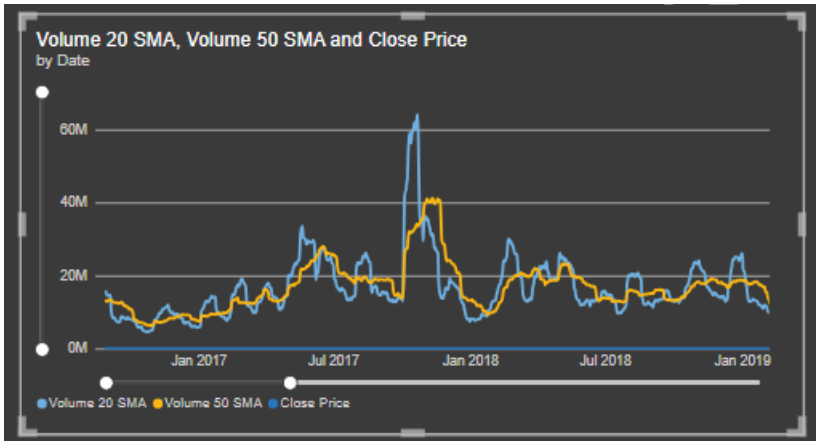
- Date Table[Date] on the X-axis
- Close Price
- 20 SMA
- 50 SMA

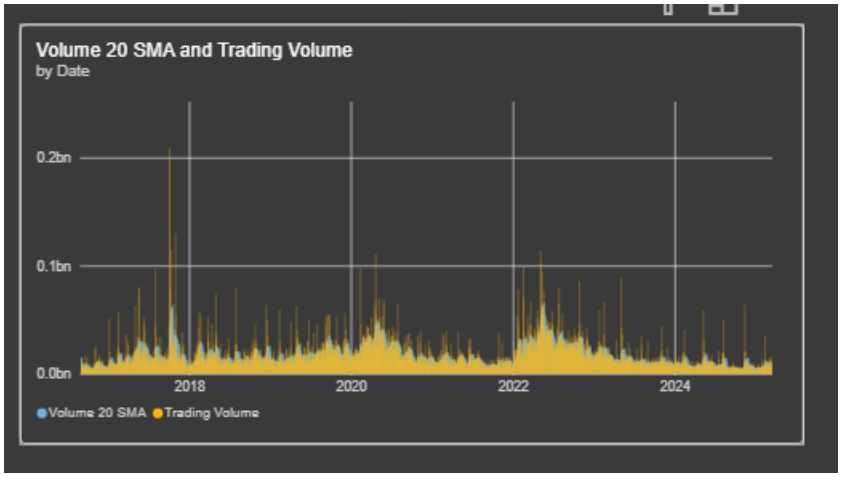
The X-axis was configured as a Continuous axis.

A Zoom Slider was also enabled to allow a specific period of the stock data to be viewed more closely.

This visualization makes it easier to observe changes and trends across the selected date range.

Interactive Stock Price Trend with Moving Averages





Trading Volume and Volatility Analysis

A separate visualization was created to analyze trading volume.

The chart contains:

- Date Table[Date] on the X-axis
- Trading Volume
- Volume 20 SMA

The visualization allows the actual trading volume to be compared with its 20-day average.

Large differences between trading volume and its moving average can help identify periods of increased trading activity.

Timeline Slicers

Two date slicers were added to make the dashboard interactive.

Slicer 1: Date Range

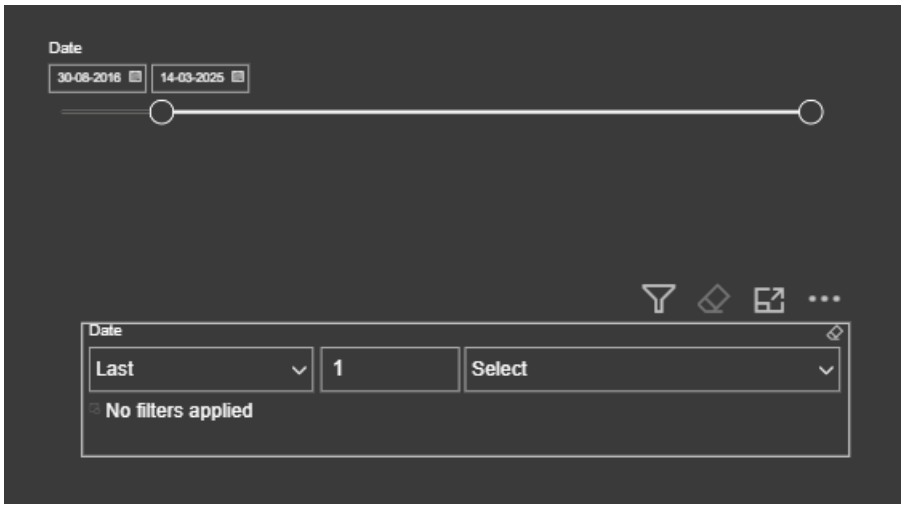
A Between date slicer was created to select a specific start and end date.

Slicer 2: Relative Date

A Relative Date slicer was created to select a recent time period.

These slicers allow users to change the period displayed in the visualizations without modifying the charts.

Timeline Slicers for Interactive Date Filtering:



Price Field Parameter

A Field Parameter named Price Parameter was created.

It contains:

- Close Price
- High Price
- Low Price
- Open Price

The parameter allows different price measures to be selected interactively from the report.

This makes the dashboard more flexible because multiple price measures can be explored without creating separate charts for each one.

Price Field Parameter



Custom Tooltip

A separate page named Stock_Tooltip was created for displaying detailed information when hovering over the main chart.

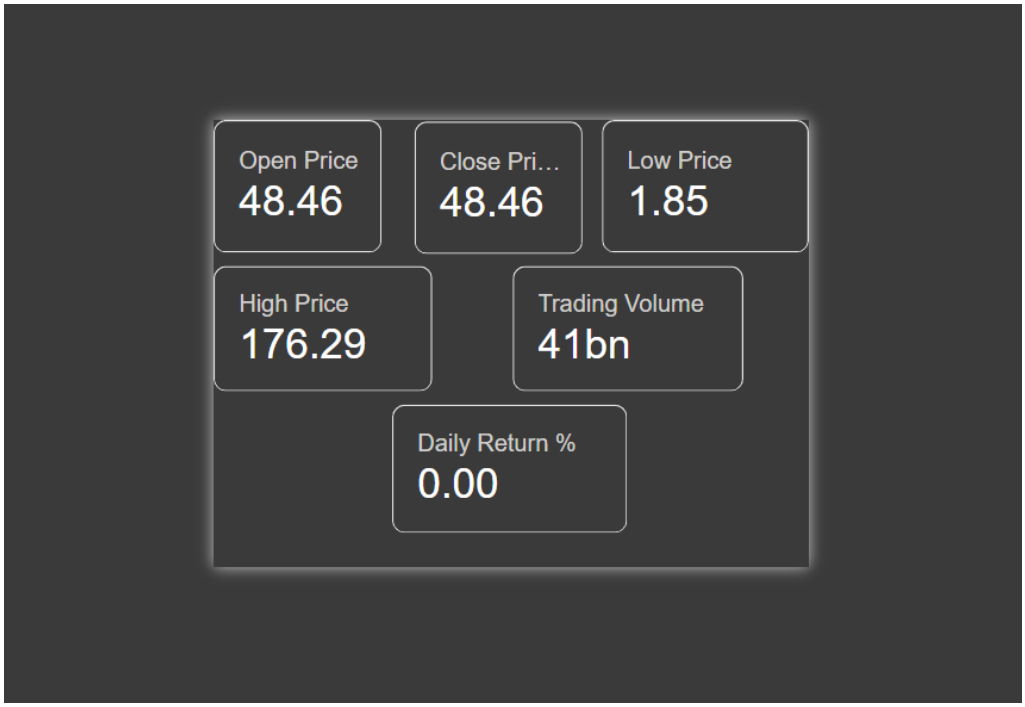
The tooltip page contains cards for:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Daily Return %

The page was configured as a Tooltip page and connected to the main price trend chart.

When the user hovers over a data point, the custom tooltip displays the corresponding stock information.

Custom Stock Tooltip Page



Custom Tooltip Displayed on the Price Trend Chart

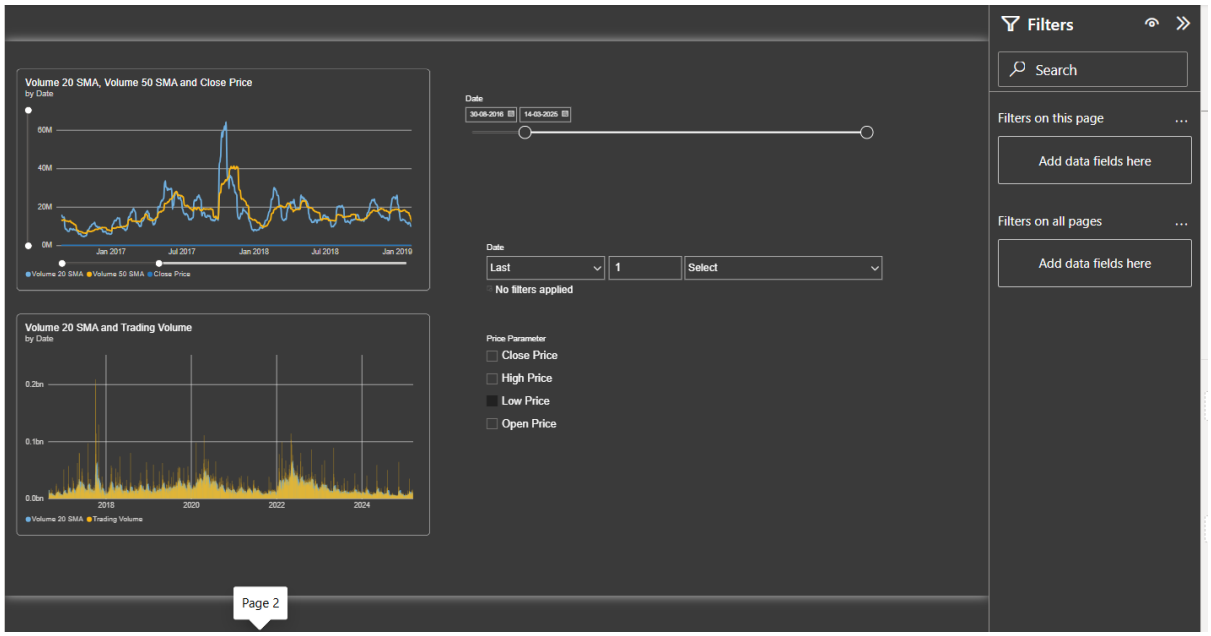


Final Dashboard

The final Power BI dashboard combines moving averages, price trends, trading volume, date filters, field parameters, and custom tooltips into an interactive stock analysis report.

The dashboard provides a simple way to explore stock performance across different time periods and compare price and volume-related measure.

Final Week 4 Power BI EDA Dashboard



Conclusion

The Week 4 Power BI EDA task provided practical experience in analyzing time-based financial data. Moving averages were used to understand short-term and long-term trends, while interactive charts and slicers improved the exploration of the dataset.

The addition of a Field Parameter and Custom Tooltip made the dashboard more interactive and informative. Overall, the task demonstrated how Power BI can be used to transform stock data into an interactive analytical dashboard.